

Exercise Series 14 (solution)

Exercise 1 Recall that if $\psi_\eta(\underline{x}) \in \{0, 1\}$ is the test function of a level α test for $H_0: \theta = \eta$, then

$\mathcal{J}(\underline{x}) = \left\{ \eta \in \Theta : \psi_\eta(\underline{x}) = 0 \right\}$ is a CI for θ with confidence coefficient $\geq 1 - \alpha$.

We derived UMP and UMPV tests for different problems in Series 11 and LRT, Wald's test, Score test in Series 12. Use those to construct the desired confidence intervals.

In many of these cases, it is difficult/impossible to invert the test to get an interval for θ . However, with Wald's test, you will always get an interval. This is because the Wald's test function is of the form

$$\psi_\eta(\underline{x}) = \begin{cases} 1 & \text{if } (\hat{\theta} - \eta)^2 I(\hat{\theta}) > \chi^2_{\alpha; 1} \\ 0 & \text{ow.} \end{cases}$$

$$\therefore \left\{ \eta : \psi_\eta(\underline{x}) = 0 \right\} = \left\{ \eta : (\hat{\theta} - \eta)^2 \leq \frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})} \right\}$$

$$= \left\{ \eta : -\sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}} \leq \eta - \hat{\theta} \leq \sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}} \right\}$$

$$= \left\{ \eta : \hat{\theta} - \sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}} \leq \eta \leq \hat{\theta} + \sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}} \right\}$$

$$= \left[\hat{\theta} - \sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}}, \hat{\theta} + \sqrt{\frac{\chi^2_{\alpha; 1}}{I(\hat{\theta})}} \right]$$

A similar construction is possible for the natural parameter of the exponential family with Rao's score test.

Exercise 2 For $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p_1)$, $Y_1, \dots, Y_m \stackrel{iid}{\sim} \text{Ber}(p_2)$, indep.

We observed that the MLE's are $\hat{p}_1 = \bar{X}$, $\hat{p}_2 = \bar{Y}$.

Also,

$$\hat{p}_1 - \hat{p}_2 \stackrel{a}{\sim} N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n} + \frac{p_2(1-p_2)}{m}\right)$$

$$\Rightarrow \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}} \stackrel{a}{\sim} N(0, 1)$$

$$\therefore \text{IP} \left[-z_{\alpha/2} \leq \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}} \leq z_{\alpha/2} \right] \approx 1 - \alpha$$

$$\Rightarrow \text{IP} \left[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}} \leq p_1 - p_2 \leq \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}} \right] \approx 1 - \alpha$$

So, $100(1-\alpha)\%$ C.I. based on Wald's test is

$$\left[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}, \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}} \right]$$

Similarly, $100(1-\alpha)\%$ C.I. based on Score test can be obtained as

$$\left[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\hat{p}_0(1-\hat{p}_0) \left(\frac{1}{n} + \frac{1}{m} \right)}, \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\hat{p}_0(1-\hat{p}_0) \left(\frac{1}{n} + \frac{1}{m} \right)} \right]$$

A one-sided CI can be obtained using the fact that

$$\text{IP} \left[\frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}} \leq z_{\alpha} \right] \approx 1 - \alpha$$

$$\Rightarrow \mathbb{P} \left[\hat{p}_1 - \hat{p}_2 - z_\alpha \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}} \leq p_1 - p_2 \right] \approx 1 - \alpha$$

$$\therefore \left[\hat{p}_1 - \hat{p}_2 - z_\alpha \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}, \infty \right) \text{ is a one-sided}$$

CI of the form $[a, \infty)$ based on Wald's test.

The same based on Rao's test would be

$$\left[\hat{p}_1 - \hat{p}_2 - z_\alpha \sqrt{\hat{p}_0(1-\hat{p}_0) \left(\frac{1}{n} + \frac{1}{m} \right)}, \infty \right)$$

Exercise 3. This is similar to the previous exercise. We have already seen that the MLEs are

$$\hat{\mu}_1 = \bar{X}, \hat{\mu}_2 = \bar{Y}, \hat{\sigma}^2 = \frac{1}{m+n} \left\{ \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{j=1}^m (y_j - \bar{y})^2 \right\}$$

$$\hat{\sigma}_0^2 = \frac{1}{m+n} \sum_{i=1}^{m+n} (z_i - \bar{z})^2 \quad [(z_1, \dots, z_{m+n}) = (x_1, \dots, x_n, y_1, \dots, y_m)]$$

We also saw that

$$\hat{\mu}_1 - \hat{\mu}_2 \stackrel{a}{\sim} N(\mu_1 - \mu_2, \sigma^2 \left(\frac{1}{n} + \frac{1}{m} \right))$$

$$\Rightarrow \frac{(\hat{\mu}_1 - \hat{\mu}_2) - (\mu_1 - \mu_2)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)}} \stackrel{a}{\sim} N(0, 1)$$

A $100(1-\alpha)\%$ two-sided CI for $\mu_1 - \mu_2$ can be obtained as

$$\left[\hat{\mu}_1 - \hat{\mu}_2 - z_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)}, \hat{\mu}_1 - \hat{\mu}_2 + z_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)} \right]$$

With score test, we get

$$\left[\hat{\mu}_1 - \hat{\mu}_2 - z_{\alpha/2} \sqrt{\hat{\sigma}_0^2 \left(\frac{1}{n} + \frac{1}{m} \right)}, \hat{\mu}_1 - \hat{\mu}_2 + z_{\alpha/2} \sqrt{\hat{\sigma}_0^2 \left(\frac{1}{n} + \frac{1}{m} \right)} \right]$$

For one-sided interval, we use

$$IP \left[-z_\alpha \leq \frac{(\hat{\mu}_1 - \hat{\mu}_2) - (\mu_1 - \mu_2)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)}} \right] \approx 1 - \alpha$$

$$\Rightarrow IP \left[\mu_1 - \mu_2 \leq \hat{\mu}_1 - \hat{\mu}_2 + z_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)} \right] \approx 1 - \alpha$$

So, a $100(1-\alpha)\%$ CI of the form $(-\infty, b]$ is

$$\left(-\infty, \hat{\mu}_1 - \hat{\mu}_2 + z_{\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{1}{m} \right)} \right]$$

Exercise 4 $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{iid}{\sim} BVN(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \ell)$

We already saw that $Z_i = X_i - Y_i \stackrel{iid}{\sim} N(\underbrace{\mu_x - \mu_y}_{\mu}, \underbrace{\sigma_x^2 + \sigma_y^2 + 2\ell}_{\sigma^2})$

So, here our problem is to find a CI for μ based on iid observation $Z_1, \dots, Z_n \sim N(\mu, \sigma^2)$, with σ^2 unknown.

An exact CI can be obtained by using

$$\frac{\bar{Z} - \mu}{S_Z / \sqrt{n}} \sim t_{n-1} \Rightarrow \left[\bar{Z} - t_{\alpha/2; n-1} \frac{S_Z}{\sqrt{n}}, \bar{Z} + t_{\alpha/2; n-1} \frac{S_Z}{\sqrt{n}} \right]$$
$$S_Z^2 = \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{Z})^2.$$

We can also construct approximate CIs by inverting Wald's / Score test, which will be similar, except S_Z^2 will be replaced by the corresponding MLE $\frac{1}{n} \sum_{i=1}^n (z_i - \bar{Z})^2$.

Exercise 5 is also similar.

Exercise 6. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, 1)$

$H_0: \mu = 0$ vs. $H_a: \mu > 0$

$$p_{\text{right}} = p_{\text{right}}(\underline{x}) = P_{H_0}(\bar{X} > \bar{x}_{0.05})$$

(a) Under H_0 , $\bar{X} \sim N(0, \frac{1}{n})$.

$$\begin{aligned} \therefore p_{\text{right}}(\underline{x}) &= P_{H_0}(\bar{X} > \bar{x}_{0.05}) \\ &= P\left(N\left(0, \frac{1}{n}\right) > \bar{x}_{0.05}\right) \\ &= P\left(N(0, 1) > \sqrt{n} \bar{x}_{0.05}\right) \\ &= 1 - \Phi(\sqrt{n} \bar{x}_{0.05}) \end{aligned}$$

$$(b) p_{\text{right}}(\underline{x}) = 1 - \Phi(\sqrt{n} \bar{X})$$

Under H_0 , $\bar{X} \sim N(0, \frac{1}{n}) \Rightarrow \sqrt{n} \bar{X} \sim N(0, 1)$

$$\begin{aligned} P(p_{\text{right}}(\underline{X}) \leq t) &= P(1 - \Phi(\sqrt{n} \bar{X}) \leq t) \\ &= P(\Phi(\sqrt{n} \bar{X}) \geq 1 - t) \\ &= P(\sqrt{n} \bar{X} \geq \Phi^{-1}(1 - t)) \\ &= P(N(0, 1) \geq \Phi^{-1}(1 - t)) \\ &= 1 - \Phi(\Phi^{-1}(1 - t)) = 1 - (1 - t) = t \end{aligned}$$

$$\Rightarrow p_{\text{right}}(\underline{x}) \sim \text{Unif}(0, 1)$$

$$\begin{aligned} (c) p_{\text{left}}(\underline{x}) &= P_{H_0}(\bar{X} < \bar{x}_{0.05}) = P_{H_0}(\sqrt{n} \bar{X} < \sqrt{n} \bar{x}_{0.05}) \\ &= P(N(0, 1) < \sqrt{n} \bar{x}_{0.05}) = \Phi(\sqrt{n} \bar{x}_{0.05}) \end{aligned}$$

$$\begin{aligned}
p_{\text{boot}}(\underline{x}) &= \mathbb{P}_{H_0}(|\bar{X}| > |\bar{x}|) \\
&= \mathbb{P}_{H_0}(\bar{X} > |\bar{x}|) + \mathbb{P}_{H_0}(\bar{X} < -|\bar{x}|) \\
&= \mathbb{P}_{H_0}(\sqrt{n}\bar{X} > \sqrt{n}|\bar{x}|) + \mathbb{P}_{H_0}(\sqrt{n}\bar{X} < -\sqrt{n}|\bar{x}|) \\
&= 1 - \Phi(\sqrt{n}|\bar{x}|) + \Phi(-\sqrt{n}|\bar{x}|) \\
&= 2\Phi(-\sqrt{n}|\bar{x}|) = 2\{1 - \Phi(\sqrt{n}|\bar{x}|)\}
\end{aligned}$$

(d) Under H_0 , $\sqrt{n}\bar{X} \sim N(0,1)$

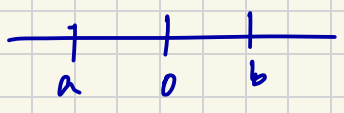
$$\begin{aligned}
\mathbb{P}(p_{\text{boot}}(\underline{X}) \leq t) &= \mathbb{P}(\Phi(\sqrt{n}\bar{X}) \leq t) \\
&= \mathbb{P}(\sqrt{n}\bar{X} \leq \Phi^{-1}(t)) \\
&= \mathbb{P}(N(0,1) \leq \Phi^{-1}(t)) = \Phi(\Phi^{-1}(t)) = t \\
\mathbb{P}(p_{\text{boot}}(\underline{X}) \leq t) &= \mathbb{P}(2\{1 - \Phi(\sqrt{n}|\bar{X}|)\} \leq t) \\
&= \mathbb{P}\left(1 - \Phi(\sqrt{n}|\bar{X}|) \leq \frac{t}{2}\right) \\
&= \mathbb{P}\left(\Phi(\sqrt{n}|\bar{X}|) \geq 1 - \frac{t}{2}\right) \\
&= \mathbb{P}\left(\sqrt{n}|\bar{X}| \geq \Phi^{-1}\left(1 - \frac{t}{2}\right)\right) \\
&= \mathbb{P}\left(\sqrt{n}\bar{X} \geq \Phi^{-1}\left(1 - \frac{t}{2}\right)\right) + \mathbb{P}\left(\sqrt{n}\bar{X} \leq -\Phi^{-1}\left(1 - \frac{t}{2}\right)\right) \\
&= 1 - \Phi\left(\Phi^{-1}\left(1 - \frac{t}{2}\right)\right) + \Phi\left(-\Phi^{-1}\left(1 - \frac{t}{2}\right)\right) \\
&= 1 - \left(1 - \frac{t}{2}\right) + 1 - \Phi\left(\Phi^{-1}\left(1 - \frac{t}{2}\right)\right) \\
&= \frac{t}{2} + 1 - \left(1 - \frac{t}{2}\right) = \frac{t}{2} + \frac{t}{2} = t \\
\therefore p_{\text{boot}}(\underline{X}) &\sim \text{Unif}(0,1), \quad p_{\text{boot}}(\underline{X}) \sim \text{Unif}(0,1).
\end{aligned}$$

Exercise 7 Define the CDF of $T(\underline{X})$ under H_0 by $F_T(\cdot)$

We assume that $F_T(\cdot)$ is continuous.

$$p_{\text{right}}(\underline{x}) = P_{H_0}(T(\underline{x}) > t_{\alpha/2}) = 1 - F_T(t_{\alpha/2})$$

$$p_{\text{left}}(\underline{x}) = P_{H_0}(T(\underline{x}) < -t_{\alpha/2}) = F_T(-t_{\alpha/2}) \quad |a| \leq |b|$$

$$\begin{aligned} p_{\text{both}}(\underline{x}) &= P_{H_0}(|T(\underline{x})| > |t_{\alpha/2}|) \\ &= 1 - P_{H_0}(|T(\underline{x})| \leq |t_{\alpha/2}|) \\ &= 1 - F_{|T|}(|t_{\alpha/2}|), \text{ where } F_{|T|}(\cdot) \text{ is the cdf} \\ &\quad \text{of } |T(\underline{x})| \text{ under } H_0. \end{aligned}$$


$$\text{Now, } p_{\text{right}}(\underline{x}) = 1 - F_T(T(\underline{x}))$$

$$p_{\text{left}}(\underline{x}) = F_T(T(\underline{x}))$$

$$p_{\text{both}}(\underline{x}) = 1 - F_{|T|}(|T(\underline{x})|)$$

Recall that if $X \sim F$, ^{conts.} then $F(X) \sim \text{Unif}(0,1)$

$$\therefore F_T(T(\underline{x})) \sim \text{Unif}(0,1), \quad F_{|T|}(|T(\underline{x})|) \sim \text{Unif}(0,1).$$

Verify that if $U \sim \text{Unif}(0,1)$, then $V = 1 - U \sim \text{Unif}(0,1)$

Thus, $1 - F_T(T(\underline{x})) \sim \text{Unif}(0,1)$, $1 - F_{|T|}(|T(\underline{x})|) \sim \text{Unif}(0,1)$.

$$\text{So, } p_{\text{right}}(\underline{x}) \sim \text{Unif}(0,1)$$

$$p_{\text{left}}(\underline{x}) \sim \text{Unif}(0,1)$$

$$p_{\text{both}}(\underline{x}) \sim \text{Unif}(0,1)$$